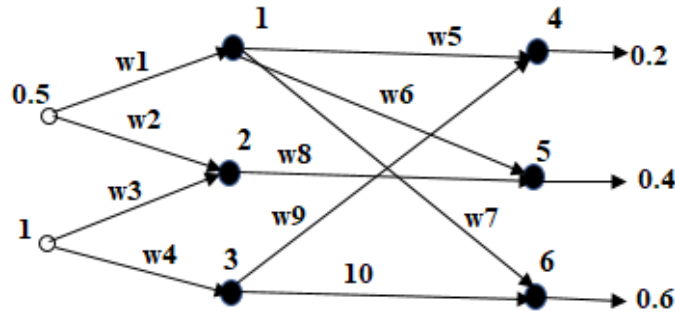
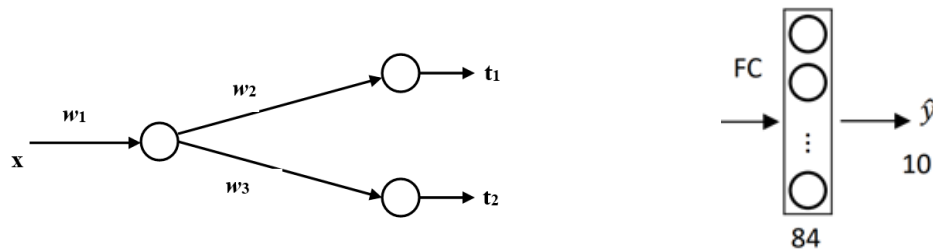


Answer all questions.

1. In the neural network shown, activation functions of the hidden layer neurons are **-logsig** (and output layer are linear (identity function)). Learning rate is 0.4, $w_1=w_2=w_3=w_5=0.2$, $w_4=w_6=w_8=w_{10}=0.1$, $w_7=0.5$, $w_9=0.4$. Find the modified error of neuron 6. Find the weight w_1 after applying backpropagation for one epoch. [Penalty for showing steps not required]. [2 +7]



2. Formulate the primal and dual problem for the linear SVM problem. The instances (2, 0.5) and (1, 1) are negative classes, (3, 1.2) is positive class. Give the formula for the margin in terms of weights and use this formula to find the margin for the above linear SVM problem. [4+3]
3. **a.** Consider two coins A and B of unknown biases θ_A and θ_B respectively. To estimate the parameters θ_A and θ_B , the following procedure is repeated 4 times: randomly choose one of the two coins (identities of the coin are not known), and perform 8 independent coin tosses with the selected coin. First time it resulted in 5 heads and 3 tails. Second time 7 heads and 1 tail. 3rd time 6 heads and 2 tails and 4th time 3 heads and 5 tails. Estimate θ_A and θ_B after one step of EM algorithm. The initial estimates are $\hat{\theta}_A = 0.6$, $\hat{\theta}_B = 0.5$. **b.** Using the d-separation criterion, show that the conditional distribution for a node x in a directed graph, conditioned on all of the nodes in the Markov blanket is independent of the remaining variables in the graph. (With the help head-tail rule and a network with the nodes inside the blanket and outside, you prove the result). [9+6]
4. The network shown below uses unipolar continuous activation functions (*logsig*). If the error function used is $E = -(t \log o + (1 - t) \log(1 - o))$, Derive the formulae for weight update of the output layer and hidden layer, Simplify the formulae. [6]



5. **a.** The figure above shows the last layers of a LeNet. How many tunable parameters are there between the last layers. Clearly explain how did you get the answer (No marks without explanation). **b.** Write short notes on a) skip connection in ResNet b) Inception layer in GoogleLeNet. [5+3+3]
6. In an SVM, $w=(1, 2)$ and $b=3$. For the point $x=(4,5)$, $y=-1$, what is the smallest slack value ξ for it to satisfy the margin constraint? [3]

7. Consider a 2D dataset where training points are on the integer grid (x_1, x_2) , and both dimensions x_1 and x_2 range from 1 to 100 (inclusive). The binary label for point (x_1, x_2) is $(-1)^{(x_1+x_2)}$. What is the label for test point (51.3, 62.1) with a 3-NN classifier? [3]
8. The table shows the confusion matrix of a classifier. Find the F1 score of the classifier.

Actual\Predicted	Positive	Negative
Positive	131	155
Negative	19	195

[3]

9. Suppose with probability $(1 - \delta)$, a PAC learner outputs a hypothesis within error ϵ of the best possible hypothesis in H . It can be shown that the lower bound on the number of training examples m sufficient for successful learning, stated in terms of $VC(H)$ (The VC dimension of hypothesis space H defined over instance space X) is

$$m \geq \frac{1}{\epsilon} (4 \log_2(2/\delta) + 8VC(H) \log_2(13/\epsilon))$$

Consider a learning problem in which X the set of real numbers, and the hypothesis space is the set of intervals $H = \{(a < x < b) | a, b \in \mathbb{R}\}$. Note that the hypothesis labels points inside the interval as positive, and negative otherwise. What is the probability that a hypothesis consistent with m examples will have error at least ϵ ? [8]

10. a. Consider a system with two states and two actions. You perform actions and observe the rewards and transitions listed below. Each step lists the current state, reward, action and resulting transition as $S_i ; R = r; a_k : S_i \rightarrow S_j$. Perform Q-learning using a learning rate of $\alpha = 0.5$ and a discount factor of $\gamma = 0.5$ for each step. The Q-table entries are initialized to zero. Also find the optimal policy after 4th step.

$S_1 \quad R = -10 \quad a_1 : S_1 \rightarrow S_1$			$S_1 \quad R = -10 \quad a_2 : S_1 \rightarrow S_2$			$S_2 \quad R = +20 \quad a_1 : S_2 \rightarrow S_1$			$S_1 \quad R = -10 \quad a_2 : S_1 \rightarrow S_2$		
Q	S_1	S_2	Q	S_1	S_2	Q	S_1	S_2	Q	S_1	S_2
a_1			a_1			a_1			a_1		
a_2			a_2			a_2			a_2		

b. Consider an undiscounted MDP having three states, (1, 2, 3), with rewards $-1, -2, 0$, respectively. State 3 is a terminal state. In states 1 and 2 there are two possible actions: a and b . The transition model is as follows:

In state 1, action a moves the agent to state 2 with probability 0.8 and makes the agent stay put with probability 0.2. In state 2, action a moves the agent to state 1 with probability 0.8 and makes the agent stay put with probability 0.2. In either state 1 or state 2, action b moves the agent to state 3 with probability 0.1 and makes the agent stay put with probability 0.9.

- (a) What can be determined qualitatively about the optimal policy in states 1 and 2?
- (b) Apply policy iteration, showing each step in full, to determine the optimal policy and the values of states 1 and 2. Assume that the initial policy has action b in both states. [6+9]